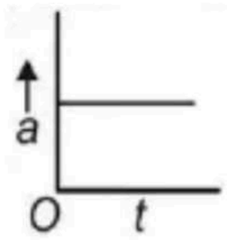
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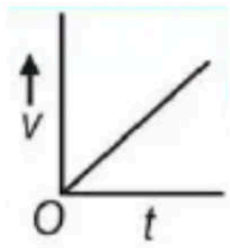
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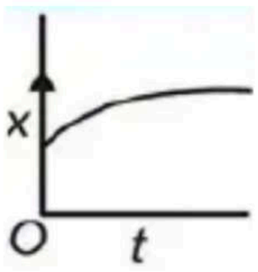
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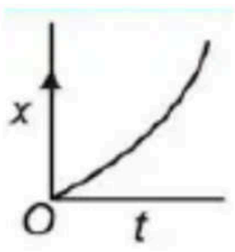
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THE PERFECT HAT-TRICK WITH ALL-INDIA RANK 1 IN JEE MAIN 2023 JEE ADVANCED 2023 AND NEET 2023

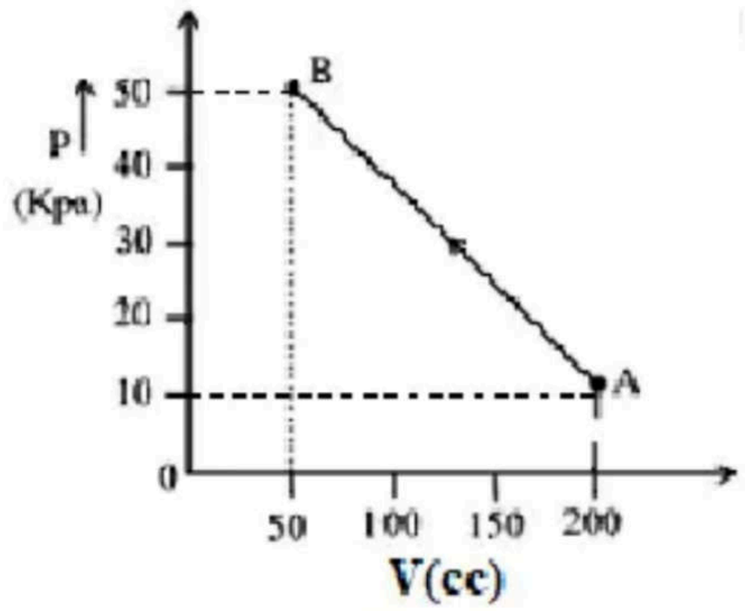

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O
t

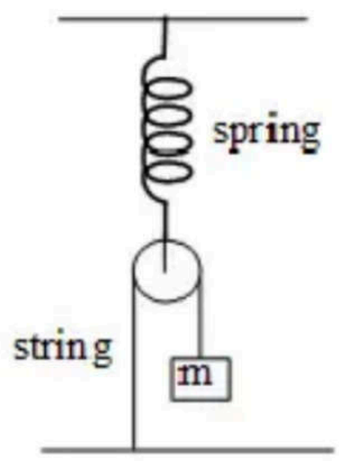

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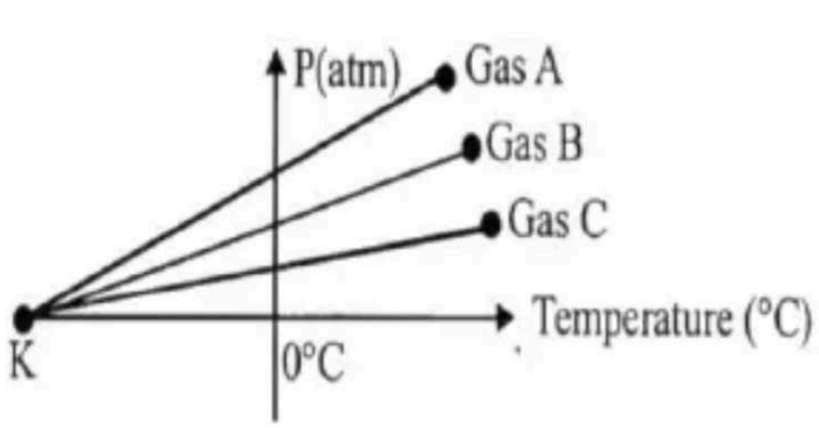

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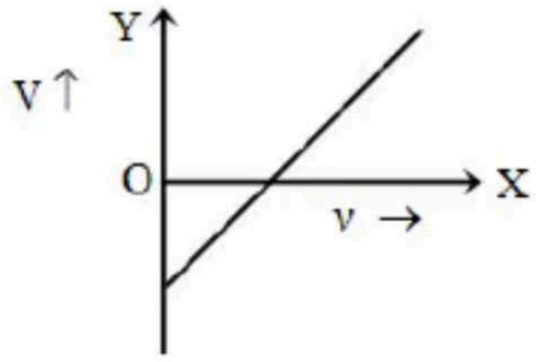

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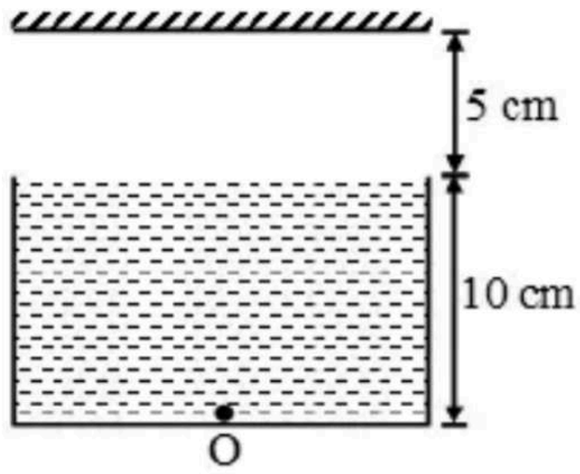
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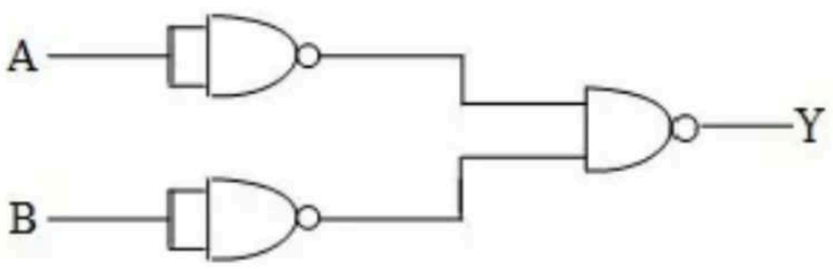

P
(Kpa)
50
40
30
20
10
0
B
A
50
100
150
200
V(cc)

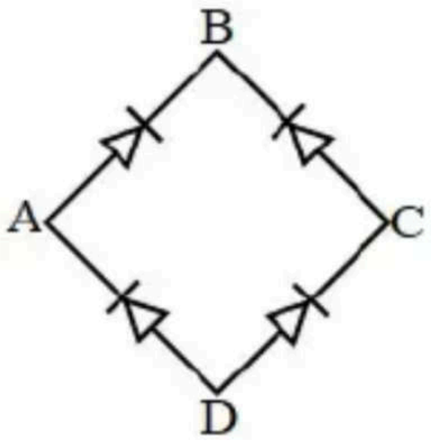

spring
string
m

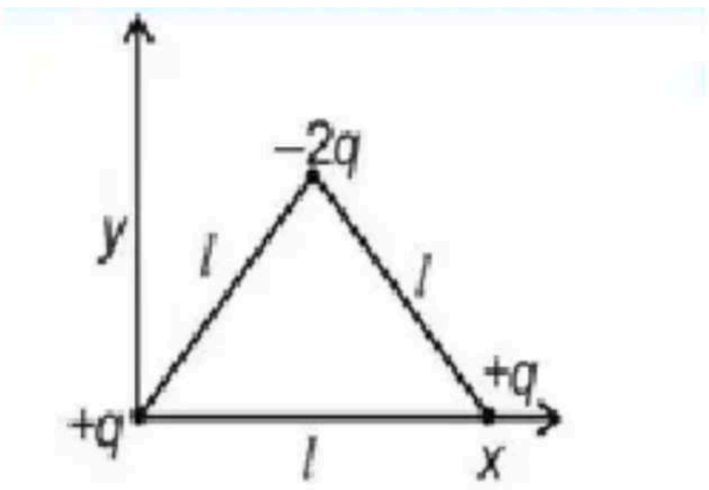

P(atm)
Gas A
Gas B
Gas C
Temperature (°C)
K
0°C


Y
V
O
X
v


5 cm
10 cm
O


A
B
Y


B
A
C
D



$(2ql)\hat{j}$ $\sqrt{3}\,ql\ \dfrac{\hat{j}\ \ \hat{i}}{\sqrt{2}}$

$\sqrt{3}\,(ql)\hat{j}$ $(ql)\dfrac{\hat{i}\ \ \hat{j}}{\sqrt{2}}$

C_1 C_2

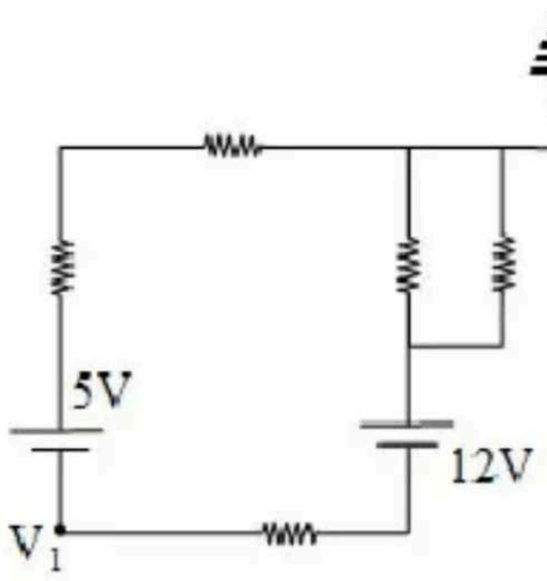
120 V 200 V

$5C_1$ $3C_2$ $3C_1$ $5C_2$

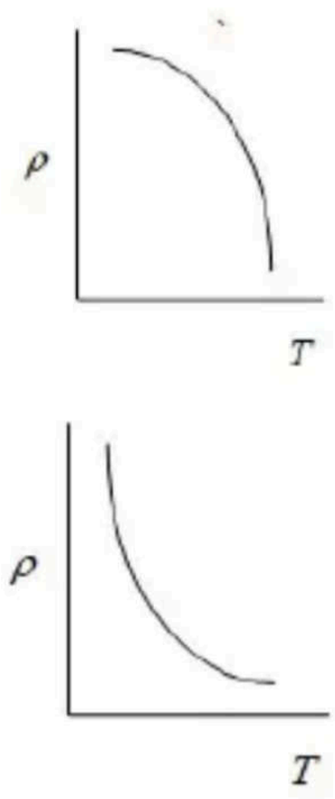
C_1 C_2 $9C_1$ $4C_2$

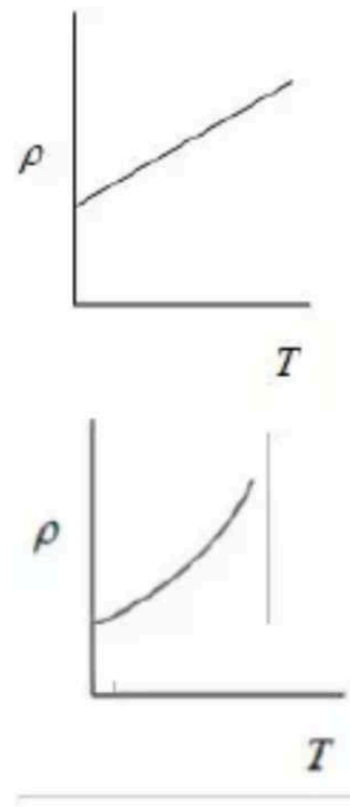
Ω

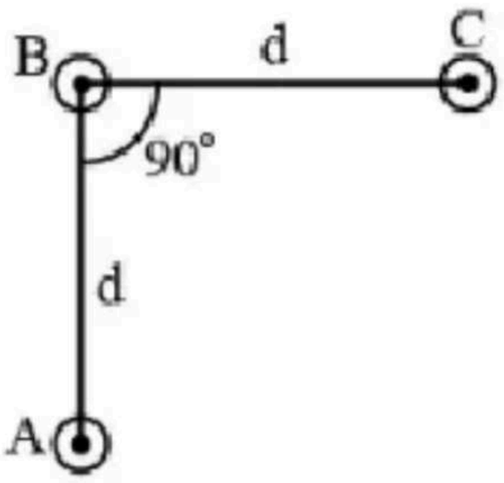
V_1



(ρ)







$\dfrac{2\ \ {}_0 i^2}{d}$ $\dfrac{\sqrt{\ }\ \mu}{\pi}$

$\dfrac{\mu}{\sqrt{\ }\ \pi}$ $\dfrac{{}_0 i^2}{2\ \ d}$

$10^{\ 4}$ A

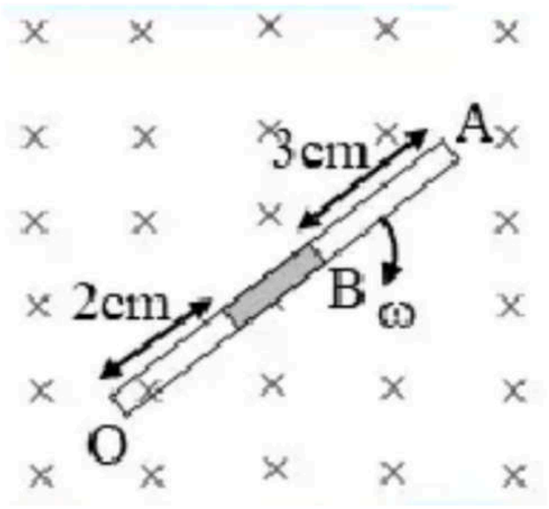
30 KΩ

0 5 V

0 10 mA

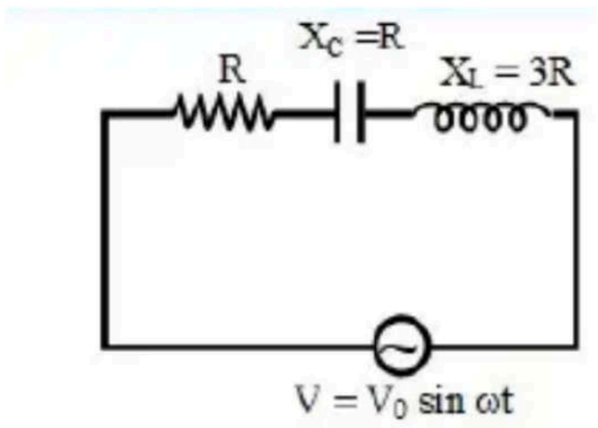
200Ω 500Ω

100Ω 10Ω



–73 C

–173 C



—

$\frac{V_0^2}{\sqrt{5}R}$

—

$\frac{V_0^2}{10R}$

$\rightarrow \quad \rightarrow$
$\times$

$\rightarrow \quad \rightarrow$

90^0

$\rightarrow \quad \rightarrow$

90^0

0.5 mm

0.5 mm

0.80 mm

0.70 mm

0.75 mm

xA^0

$(in\ A^0)$

–

$\frac{5}{27}x$

$\frac{27}{5}x$

$\frac{5}{36}x$

I_3

I_3

sp^3d

$AB_2(g)$

$AB_2(g)$ $AB(g) + B(g)$

AB_2

49. Assertion (A): pH of $^{-}$ M HCl is less than 7 at 25°C.
Reason (R): At very low concentration of HCl, contribution of $^{+}$ from water is considerable.
(1) Both A and R are true, and R is the correct explanation of A
(2) Both A and R are true, but R is not the correct explanation of A
(3) A is true, but R is false
(4) A is false, but R is true

50. Match Column I with Column – II and choose the correct option

Column-I Sign of ΔH and ΔS		Column-II nature of reaction	
A.	- & -	I.	Spontaneous only at low temperature
B.	- & +	II.	Spontaneous only at high temperature
C.	+ & +	III.	Spontaneous at all temperature
D.	+ & -	IV.	Non-spontaneous at all temperature

(1) A – II, B – I, C – IV, D – III
(2) A – III, B – IV, C – II, D – I
(3) A – I, B – III, C – II, D – IV
(4) A – IV, B – III, C – II, D – I

51. If 0.01M solution of an electrolyte has a resistance of 40 ohms in a cell having a cell constant of $^{-}$ then its molar conductance in $^{-}$ $^{-}$ will be
(1) (2)
(3) (4) 10

52. The orbit and orbital angular momentum of an electron are $\frac{3h}{2\pi}$ and $\sqrt{\frac{3}{2}}.\frac{h}{\pi}$ respectively. The number of radial and angular nodes for the orbital in which the electron is present are respectively
(1) 0, 2 (2) 2, 0
(3) 1, 2 (4) 2, 2

53. 10 moles of and 15 moles of were passed over a catalyst to produce 8 moles of SO_3. The ratio of moles of and SO_3 in the mixture is
(1) – (2) –
(3) – (4) –

54. Given below are two statements:
Statement – I : Boiling point of H_2O is higher than boiling point of HF.
Statement – II : H-F ---- H hydrogen bonding stronger than H – O --- H hydrogen bonding.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both the statements I and II are correct
(2) Both the statements I and II are incorrect
(3) Only statement I – is correct, statement – II is incorrect
(4) Only statement I – is incorrect, statement – II is correct

55. The enthalpy of solution of anhydrous $CuSO_4$ is –15.9kCal and that of $CuSO_4.5H_2O$ is 2.8kCal. Enthalpy of hydration of anhydrous $CuSO_4$ is
(1) 18.7kCal (2) -18.7kCal
(3) 13.1kCal (4) -13.1kCal

56. Consider the following reactions:
(i) $C_2O_4^{2-} \rightarrow CO_2$ (ii) $SO_4^{2-} \rightarrow SO_3^{2-}$
(iii) $MnO_4^{2-} \rightarrow MnO_4^{-}$ (iv) $Fe^{3+} \rightarrow Fe^{2+}$
Choose the correct answer:
(1) (i) & (ii) show oxidation
(2) (iii) & (iv) show reduction
(3) (i) & (iii) show oxidation
(4) (iii) & (iv) show oxidation

57. K_a for HCN is × $^{-}$ .For maintaining a constant of 9.6020, the volume of 5M KCN solution required to be added to 10ml of 2M HCN solution is
(1) 2 ml (2) 4 ml
(3) 8 ml (4) 16 ml

58. In the formation of sulphur trioxide by contact process, $2SO_2 + O_2 \rightarrow 2SO_3$, the rate of reaction was measured as $-\frac{d[O_2]}{dt} = 2.5 \times 10^{-4} \text{mol L}^{-1}\text{s}^{-1}$. The rate of reaction expressed in terms of SO_3 will be

(1) $-5.0 \times 10^{-4} \text{mol L}^{-1}\text{s}^{-1}$

(2) $-1.25 \times 10^{-4} \text{mol L}^{-1}\text{s}^{-1}$

(3) $2.5 \times 10^{-4} \text{mol L}^{-1}\text{s}^{-1}$

(4) $5.00 \times 10^{-4} \text{mol L}^{-1}\text{s}^{-1}$

59. In a 0.2 molal aqueous solution of a weak acid HX, the degree of ionization is 0.3. Taking K_f for water as 1.85, the freezing point of the solution will be nearest to

(1) -0.360ºC (2) -0.260ºC

(3) +0.480ºC (4) -0.480ºC

60. Lyophobic colloid among the following is

(1) Starch (2) Gelatin

(3) $Fe(OH)_3$ (4) Gum

61. N_2 and O_2 are converted to mono cations N_2^+ and O_2^+ respectively .The wrong statement with respect to given changes in the following is

(1) The N-N bond weakens

(2) The O-O bond order increases

(3) The magnetic moment of O_2 decreases

(4) N_2^+ becomes diamagnetic

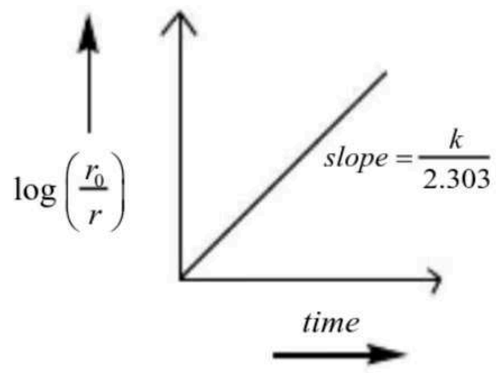
62. An aqueous solution of 6.3 g oxalic acid dihydrate is made up to 250 ml. The volume of 0.1 N NaOH required to completely neutralize 10 ml of this solution is

(1) 40 ml (2) 20 ml

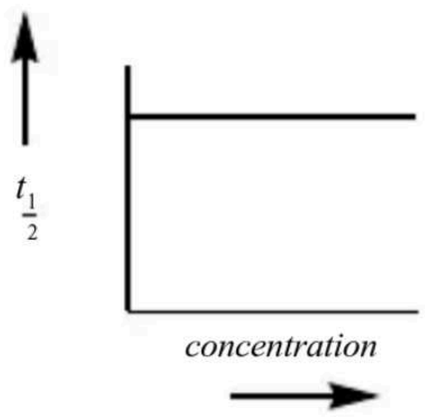
(3) 10 ml (4) 4 ml

63. Which of the following plot does not represent first order reaction? (r_o, r are concentrations of reactant initially and at time t)

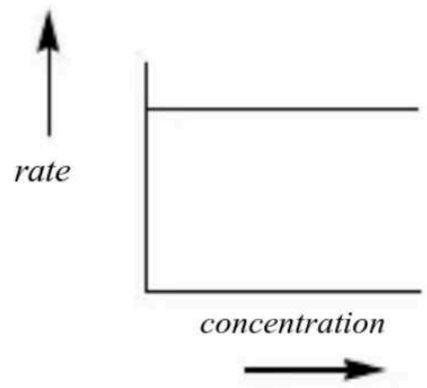
(1)



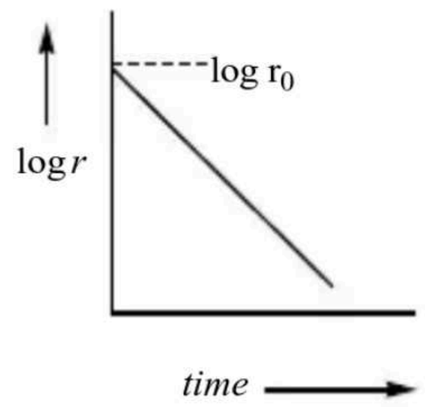
(2)



(3)



(4)



64. Incorrect statements among the following are

(A) HNO_2 disproportionates in acidic medium to release a gaseous nitrogen compound which is acidic in nature.

(B) All trihalides of nitrogen are stable where as pentahalides of nitrogen do not exist.

(C) Thermal stability of hydrides of chalcogens increases down the group.

(D) Among hexahalides of group 16 elements only hexafluorides are stable.

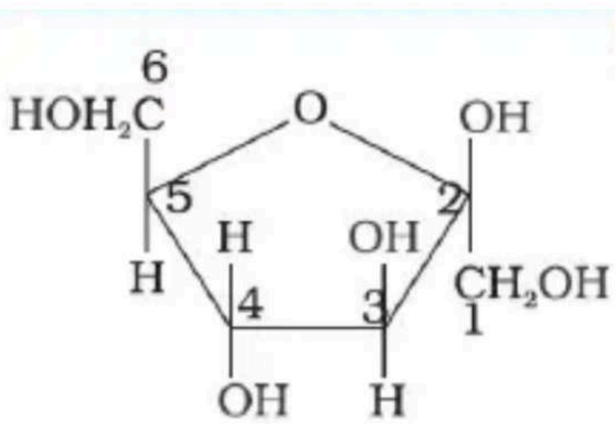
Choose the correct answer from the options given below.

(1) B,C,D only (2) A, B, C,D

(3) A,B,C only (4) A,C only

65. Given below are two statements:
Statement – I : Most of the reactions of fluorine are exothermic due to small and strong bond formed by it with other elements.
Statement – II : $XeOF_4$ is a blue solid with trigonal bipyramidal structure.
In the light of the above statements, choose the correct answer from the options given below:
(1) Both the statements I and II are correct
(2) Both the statements I and II are incorrect
(3) Only statement I – is correct, statement – II is incorrect
(4) Only statement I – is incorrect, statement – II is correct

66.



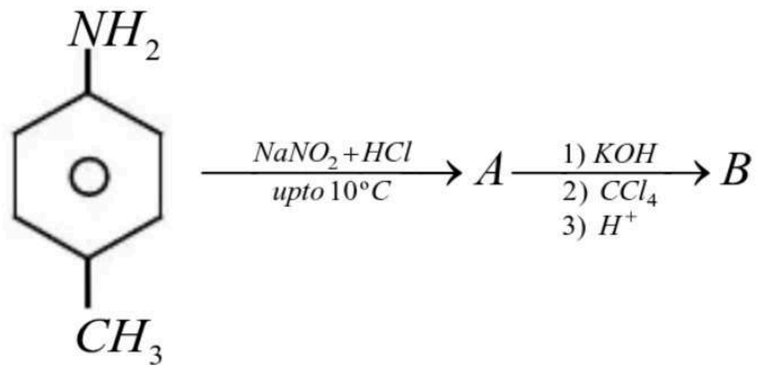
Given cyclic structure corresponds to
(1) $\alpha - D - (+) - Glucopyranose$
(2) $\beta - D - (+) - Glucopyranose$
(3) $\beta - D - (-) - Fructofuranose$
(4) $\alpha - D - (-) - Fructofuranose$

67. The reduction potential of hydrogen half-cell will be negative if
(1) $p(H_2) = 1 atm \text{ and } [H^+] = 1M$
(2) $p(H_2) = 1 atm \text{ and } [H^+] = 2M$
(3) $p(H_2) = 2 atm \text{ and } [H^+] = 1M$
(4) $p(H_2) = 2 atm \text{ and } [H^+] = 2M$

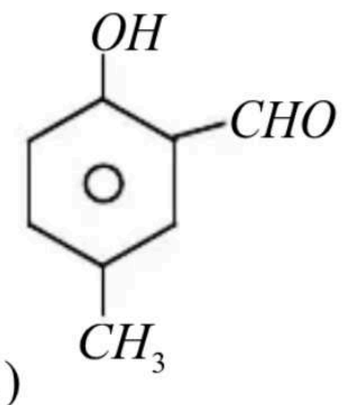
68. Hybridisation of which of the following complex is not correctly matched
(1) $[MnBr_4]^{-2} - sp^3$
(2) $[Co(OX)_3]^{3-} - sp^3d^2$
(3) $[Ni(CN)_4]^{-2} - dsp^2$
(4) $[Fe(NH_3)_6]^{2+} - d^2sp^3$

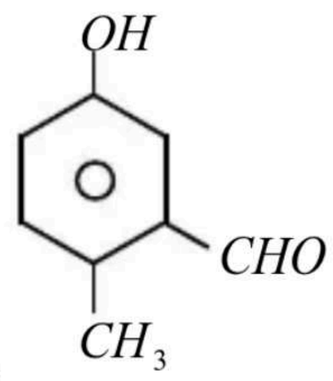
69. Identify correct statements
A) Primary aromatic amines donot give diazonium salts when treated with $NaNO_2$ in acidic condition at low temperature.
B) Aliphatic and aromatic primary amines on heating with $CHCl_3$ and alcoholic KOH form their respective carbyl amines.
C) Para amino benzene sulphonic acid is also called sulphanalic acid and exist as zwitter ion in solution.
D) Tertiary amines reacts with Hinsberg reagent easily
Choose the correct answer from the options given below
(1) B & D only (2) A & B only
(3) C & D olnly (4) B & C only

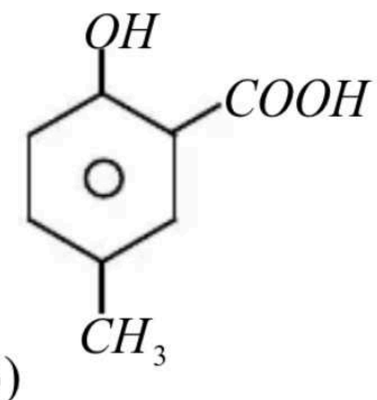
70.

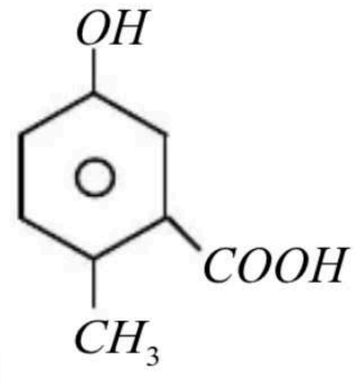


. The final major product 'B' is

(1) 

(2) 

(3) 

(4) 

71. A liquid was mixed with ethanol and a drop of conc. H_2SO_4 was added. A compound with fruity smell was formed. The liquid was
(1) CH_3OH
(2) $HCHO$
(3) $CH_3 - CO - CH_3$
(4) $CH_3 - COOH$

72. Identify the product and its colour, when MnO_2 is fused with solid KOH in presence of KNO_3 in air or O_2

(1) $KMnO_4$ & purple

(2) $KMnO_4$ & dark green

(3) K_2MnO_4 & purple

(4) K_2MnO_4 & dark green

73. Which of the following is incorrect about cerium.

(1) The oxidation states shown by Ce are + 3 and + 4

(2) The stability of Ce^{+3} is more than Ce^{+4}

(3) Ce^{+4} ion act as an oxidising agent in solution

(4) Ce^{+4} is not known in solution

74. The complex $\left[Co\left(NH_3\right)_5\left(NO_2\right)\right]Cl_2$ and $\left[Co\left(NH_3\right)_5\left(ONO\right)\right]Cl_2$ are examples of

(1) Linkage isomers

(2) Co-ordination isomers

(3) Geometrical isomers

(4) Optical isomers.

75. Assertion (A) : 1 – butene on reaction with HBr in presence of peroxide produce 1 – bromo butane.

Reason (R) : It involves free radical addition reaction mechanism and follows anti Markovnikov's addition.

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

76. Incorrect statement among the following is

(1) IP_1 of Al < IP_1 of Mg

(2) IP_2 of Mg > IP_2 of Na

(3) IP_1 of Na < IP_1 of Mg

(4) IP_3 of Mg > IP_3 of Al

77. In kjeldhal's estimation of nitrogen, the NH_3 evolved from 0.5 gm of organic compound neutralised by 10 ml of 1 M H_2SO_4 solution. Then % of nitrogen in the compound is

(1) 28 (2) 42

(3) 14 (4) 56

78. The most stable carbocation among the following is

(1) $CH_3-\overset{\oplus}{C}H_2$ (2) $H_3C-\underset{CH_3}{\overset{\oplus}{C}}-CH_3$

(3) $CH_3-\overset{\oplus}{C}H-CH_3$ (4) $\overset{\oplus}{C}H_3$

79. $CD_2=CH-CH_2-Br$ is subjected to SN^1 and SN^2 reactions separatly with nucleophile (Nu). Which of the following statements is correct?

(1) SN^1 gives one product & SN^2 gives two products

(2) SN^1 gives two products & SN^2 gives one product

(3) Both SN^1 & SN^2 gives only one product

(4) Both SN^1 & SN^2 give two products

80. The reaction of toluene with one mole of Cl_2 in presence of $FeCl_3$ produce 'X'. While the reaction in presence of sun light gives 'Y'. Then 'X' and 'Y' respectively are

(1) X = Benzyl chloride Y = O & P - Chlorotoulene

(2) X = O & P chlorotoluene Y = Benzyl chloride

(3) X = O & P chlorotoluene Y = Chloro toulene

(4) X = m - chlorotoluene Y = Benzyl chloride

81. In the reaction, $C_6H_5-CN\xrightarrow[2)\ H_3O^+/\Delta]{1)\ SnCl_2+HCl}$ product. The modification of functional group and change in hybridisation of functional group carbon respectively are

(1) – CN into $-CH_2-NH_2$ & sp^2 to sp^3

(2) – CN into $-CO-NH_2$ & sp to sp^2

(3) – CN into $-CHO$ & sp to sp^2

(4) – CN into $-CHO$ & sp to sp^3

82. $C_2H_5OH + Na \longrightarrow \frac{1}{2} X + H_2$

$(CH_3)_3 COH + PCl_3 \longrightarrow Y + H_3PO_3$

$X + Y \longrightarrow Z$; IUPAC name of 'Z' is

(1) Isobutane

(2) 2-ethoxy-2-methyl propane

(3) 2-methyl-1- propene

(4) Tertiary butyl ethyl ether

83. The end product (C) in the following sequence of reaction is

$$CH \equiv CH \xrightarrow[20\%\ H_2SO_4]{1\%\ HgSO_4} A \xrightarrow[2)\ H_3O^+]{1)\ CH_3MgBr} B \xrightarrow[CS_2]{PCC} C$$

(1) Acetic acid (2) Isopropyl alcohol

(3) Acetone (4) Ethanol

84. The order of nucleophlic substitution reaction (SN^1) in the following are:

(A) $CH_3CH_2\overset{\overset{Br}{|}}{C}HCH_3$

(B) $CH_3CH_2\overset{\overset{I}{|}}{C}HCH_3$

(C) $CH_3 - \overset{\overset{I}{|}}{\underset{\underset{CH_3}{|}}{C}} - CH_3$

(D) $CH_3 - \overset{\overset{Br}{|}}{\underset{\underset{CH_3}{|}}{C}} - CH_3$

(1) C > D > B > A (2) A > B > C > D

(3) B > A > D > C (4) C > B > D > A

85. An organic compound C_3H_6O, does not give to precipitate with 2, 4 – DNP and does not react with metallic sodium. It could be

(1) $CH_3 - CH_2 - CHO$

(2) $CH_3 - CO - CH_3$

(3) $CH_2 = CH - CH_2 - OH$

(4) $CH_2 = CH - O - CH_3$

86. Consider the following statements

A) Gallium is used as pyrometric liquid because of large difference in its MP & BP

B) Borax is used as a flux for soldering of metals

C) Diamond is a good conductor of electricity

D) Graphite is used in making refractory crucibles

E) C_{60} is the pure form of carbon because it has dangling bond.

Identify the correct statements

(1) A, B, D & E only

(2) A, B, C & E only

(3) A, B & D only

(4) A, C, D & E only

87. Given below are two statements:

Statement – I : Lucas reagent is a mixture of anhyd $ZnCl_2$ and conc. HCl

Statement – II : Primary alcohols produces precipitate with Lucas reagent immediately

In the light of the above statements, choose the correct answer from the options given below:

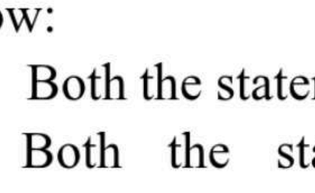
(1) Both the statements I and II are correct

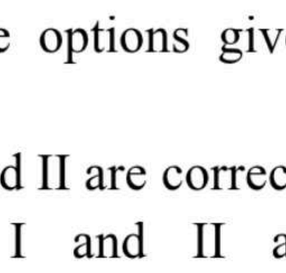
(2) Both the statements I and II are incorrect

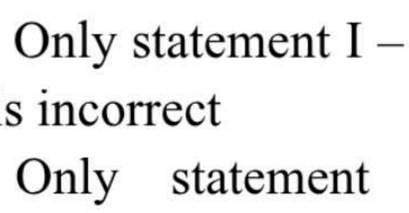
(3) Only statement I – is correct, statement – II is incorrect

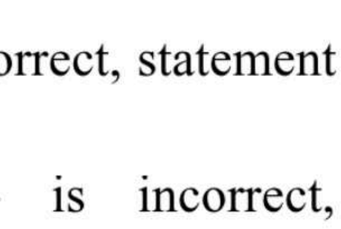
(4) Only statement I – is incorrect, statement – II is correct

88. Identify the correct order of acidic strength of the following compounds.

A) 

B) 

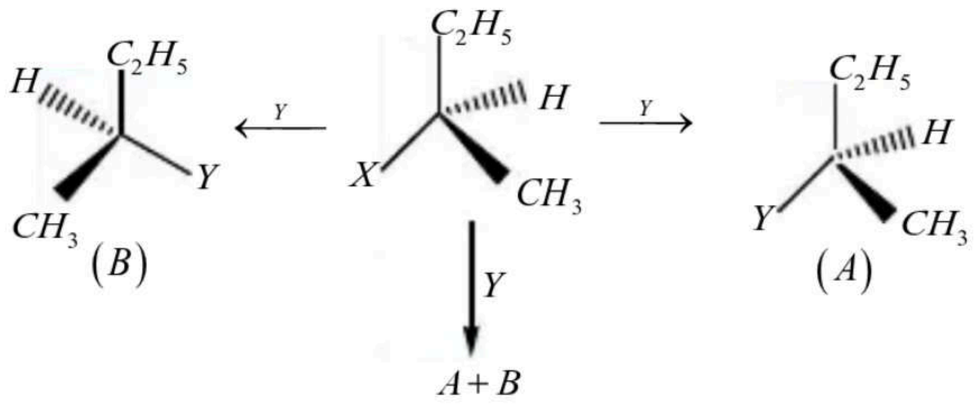
C) 

D) 

(1) C > B > D > A (2) C > B > A > D

(3) B > C > A > D (4) C > B > D > A

89.



For the reaction scheme given above, some statements are given, indentify the statements are True (T) (or) False (F) and choose the correct option.

i) If 'A' is the only compound obtained, the process is called retention of configuration

ii) It 'B' is the only compound obtained, the process is called inversion of configuration

iii) If a 50 : 50 mixture of the above two is obtained then the process is called racemisation

iv) If a 50 : 50 mixutre of A & B is obtained, then it is racemization, and the product is optically active.

(1) T T T F (2) T F T F

(3) T F T T (4) T F F T

90. Match the following columns

Column – I		Column – II	
A.	Amide	I.	Hoffmann bromamide reaction
B.	Alkene	II.	Carbyl amine reaction
C.	1° amine	III.	Bayer's reagent
D.	Phenol	IV.	Neutral $FeCl_3$

(1) A – I, B – III, C – III, D – IV

(2) A – II, B – I, C – III, D – IV

(3) A – I, B – III, C – II, D – IV

(4) A – II, B –IV, C – I, D – III

BIOLOGY

91. In RNAi-based nematode-resistant plants, production of both sense and antisense RNA leads to nematode death mainly because:

(1) dsRNA damages nematode DNA

(2) dsRNA triggers degradation of nematode mRNA, blocking protein synthesis

(3) Antisense RNA inhibits transcription

(4) Plant toxins kill the nematode

92. Failure of cytokinesis after karyokinesis results in a multinucleate condition called syncytium. Which of the following is a classical example of this condition?

(1) Endosperm of wheat

(2) Root tip cells

(3) Liquid endosperm of coconut

(4) Pollen grains of angiosperms

93. Read the following statements and choose the option which represents correct combination.

A. *Obelia* exhibits metagenesis.

B. Ostrich and *Ornithorhincus* are flightless birds.

C. All tetrapods have four chambered heart.

Select the correct option

	A	B	C
(1)	True	True	False
(2)	True	False	True
(3)	False	False	True
(4)	True	False	False

94. A newly discovered flowering plant species from the Western Ghats is to be given a scientific name that will be universally accepted and must follow internationally agreed rules. Which code should the taxonomist follow?

(1) ICBacN (2) ICZN

(3) ICTV (4) ICBN

95. Find the correctly matched in relation to vertebrates.

(1) *Aptenodytes*– Respiration supplemented by air sacs

(2) *Chemeleon* – Four chambered heart and a poikilothermic animal.

(3) *Icthyophis* - Sexes are separate, two chambered heart

(4) Fighting fish – 6-15 pairs of gills which are covered by operculum

96. Match Column I with Column II and choose the correct option:

Column – I (Stage/Structure)		Column – II (Description)	
A.	Prophase	I.	Sister chromatids separate
B.	Metaphase	II.	Nuclear envelope disappears
C.	Anaphase	III.	Kinetochores attach to spindle fibres
D.	Telophase	IV.	Reformation of nuclear membrane

(1) A – II, B – III, C – I, D – IV

(2) A – III, B – II, C – I, D – IV

(3) A – II, B – I, C – III, D – IV

(4) A – IV, B – III, C – I, D – II

97. Match the Column I with Column II and find the correct option.

Column I		Column II	
A	*Asterias*	1	Devil fish
B	*Clarias*	2	Star fish
C	*Octopus*	3	Cat fish
D	*Aurelia*	4	Jelly fish

(1) A-1, B-4, C-3, D-2

(2) A-2, B-3, C-1, D-4

(3) A-4, B-1, C-3, D-2

(4)A-2, B-3, C-4, D-1

98. An organism found in stagnant fresh water lacks a cell wall, possesses a protein-rich pellicle, shows photosynthesis in light but behaves as a heterotroph in darkness. This organism is:

(1) *Gonyaulax* (2) Diatoms

(3) *Euglena* (4) *Spirogyra*

99. Assertion (A): Anaphase II of meiosis is similar to anaphase of mitosis.

Reason (R): In anaphase II, sister chromatids separate due to division of centromeres and move to opposite poles.

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

100. Which of the following features are same in Coelenterata and Ctenophora ?

(1) Presence of segmentation and complete gut.

(2) Radial symmetry and diploblastic nature

(3) Dorso-ventrally flattened body and coelom

(4) Notochord and presence of false coelom

101. In mosses, the sporophyte is differentiated into foot, seta, and capsule, spores are formed after meiosis, and vegetative reproduction occurs by fragmentation and budding in secondary protonema. Which of the following is a correct example of such a moss?

(1) *Marchantia* (2) *Riccia*

(3) *Funaria* (4) *Chara*

102. In gymnosperms, the male and female gametophytes are not free-living and remain within the sporangia on the sporophyte. Which of the following correctly represents the male gametophytic generation in gymnosperms?

(1) Multicellular structure bearing antheridia

(2) Independent haploid plant body

(3) Free-living protonema

(4) Highly reduced pollen grain formed within microsporangium

103. Assertion (A): A flower is a modified shoot.
Reason (R): During flower formation, the shoot apical meristem is converted into a floral meristem, internodes do not elongate, and the axis becomes condensed.
(1) Both A and R are true, and R is the correct explanation of A
(2) Both A and R are true, but R is not the correct explanation of A
(3) A is true, but R is false
(4) A is false, but R is true

104. Read the following and choose the incorrect set with reference to cartilage.
a) It is solid with non pliable matrix
b) Epiglottis is a thin elastic cartilaginous flap.
c) Collagen fibres are absent
d) Matrix has chondroitin salts
e) It is a specialised connective tissue.
(1) a, c, d (2) All except b
(3) a, c only (4) a, b, c only

105. Given below are two statements :
Statement I: Muscles with intercalated discs are involuntary in their function.
Statement II: Contractility and elasticity are special properties of both muscles and neurons.
In the light of the above statements, choose the correct answer from the options given below :
(1) Statement I is true, and statement II is false.
(2) Both statement I and II are true.
(3) Statement I is false and statement II is true.
(4) Both statement I and statement, II are false

106. Which of the following structures are present only in male cockroach ?
(1) Tegmina and Malpighian tubules
(2) Anal cerci and phallic glands
(3) Caudal style and Titillator
(4) Anal style and anterior aorta

107. Which of the following statements are correct?
I. *Euglena* shows mixotrophic mode of nutrition.
II. Brown algae store food as laminarin and mannitol.
III. Dinoflagellates possess a silica cell wall.
IV. High BOD indicates high organic pollution.
V. Pollen grain represents the male gametophyte in *Brassica*.
Options:
(1) I, II and IV only (2) I, II, IV and V
(3) II, III and IV (4) I, III and V

108. Correct about conus-arteriosus in frog
(1) It is a sac like structure present on ventral side of heart
(2) It is a triangular structure joins to right atrium
(3) It receives blood through major veins called venacava
(4) It is a special venous connection between liver and intestine

109. Match Column I with Column II and choose the correct option:

Column – I		Column – II	
A.	Marginal placentation	I.	Mustard
B.	Axile placentation	II.	China rose
C.	Parietal placentation	III.	Pea
D.	Free central placentation	IV.	*Dianthus*

(1) A – III, B – II, C – I, D – IV
(2) A – I, B – III, C – II, D – IV
(3) A – III, B – I, C – II, D – IV
(4) A – IV, B – II, C – I, D – III

110. Choose the correct statements about amoebiasis.
(I) Pathogen for this disease releases haemozoin and ruptures RBC
(II) Its caused by a protozoan parasite which lives in large intestine of human
(III) *Anapheles* mosquito act as mechanical carriers for the parasite.
(IV) It is caused due to roundworm and blocks intestinal passage.
(1) (I) and (II) (2) (II) and (III)
(3) (I), (III) and (IV) (4) Only II

111. Select the incorrect statement among the following :
(1) Morphine is a very effective sedative and pain killer.
(2) Extracts of *Atropa* and *Datura* are hallucinogenic
(3) Nicotine is an alkaloid obtained from *Erythroxylum coca*
(4) Marijuana and hashish are cannabinoids

112. The multicellular or unicellular outgrowths present on the epidermis of the stem, which help in reducing water loss due to transpiration and may be secretory, are called:
(1) Root hairs (2) Stomata
(3) Trichomes (4) Subsidiary cells

113. The pollen tube enters the ovule through the micropyle and then enters one of the synergids. Which structure present in the synergid guides the entry of the pollen tube?
(1) Chalazal region
(2) Nucellus
(3) Filiform apparatus
(4) Antipodal cells

114. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R :
Assertion: In passive immunization, preformed antibodies are given to the patients.
Reason: Polio vaccine contains preformed antibodies to generate memory cells in our body.
In the light of the above statements, choose the most appropriate answer from the options given below :
(1) Both assertion and reason are true and Reason is the correct explanation of assertion.
(2) Both assertion and reason are true but reason is not the correct explanation of assertion.
(3) Assertion is true but reason is false.
(4) Assertion is false but reason is true

115. Select the correct combination
(1) Competition – Phytoplankton and Zooplankton.
(2) Predation – *Plasmodium* and human.
(3) Commensalism –The cattle egret and grazing cattle.
(4) Parasitism – *Cathamalus* and *Balanus*

116. Which of the following mechanisms genetically prevents self-pollination by inhibiting pollen germination or pollen tube growth in the pistil?
(1) Dichogamy
(2) Herkogamy
(3) Unisexuality
(4) Self-incompatibility

117. "In 1981 the "r" value for human population in India is 0.0205" this statement represents.
(1) Indian population is declining
(2) Birth rate is higher than death rate
(3) Birth rate is lower than death rate.
(4) Birth rate is equal to death rate

118. The outer and inner surfaces of the plasma membrane of human RBC differ in composition. Which feature best reflects this asymmetry?
(1) Uniform distribution of phospholipids
(2) Presence of cholesterol on both surfaces
(3) Carbohydrate chains present only on the outer surface
(4) Equal protein orientation on both sides

119. A sudden rupture of lysosomal membranes inside a cell will immediately lead to:
(1) Increased ATP synthesis
(2) Protein synthesis
(3) Inhibition of respiration
(4) Autolysis of cellular components

120. Choose incorrect statement about flow of energy in an ecosystem
(1) It is unidirectional flow from sun to consumers through producers.
(2) Sun is the only source of energy for all ecosystems with few exemptions.
(3) Plants capture only 2 to 10% of PAR.
(4) The energy trapped in an organism remains in it forever

121. Which of the following is incorrect combination ?
(1) Detritus - dead remains and fecal matter
(2) Detritivore – earth worm
(3) Leaching – converting inorganic matter to organic matter.
(4) Humus - reservoir of nutrients

122. A defect in Golgi-mediated tagging of enzymes will most directly affect which organelle of endomembrane system ?
(1) Mitochondrion (2) Ribosome
(3) Lysosome (4) Nucleus

123. Which of the following ecological pyramid is similar in shape with energy pyramid?
(1) Pyramid of biomass in sea
(2) Pyramid of numbers in a big tree ecosystem with number of insects feeding on tree and number of birds depending on insects.
(3) Pyramid of number in grass land ecosystem.
(4) Pyramid of biomass in related to phytoplankton and zooplankton

124. Which of the following correctly represents the structural relationship among nitrogen bases, nucleosides, nucleotides, and nucleic acids?
(1) Nitrogen base+phosphate → nucleoside
(2) Nitrogen base + sugar → nucleotide
(3) Nucleoside+ nucleoside → nucleic acid
(4) Nitrogen base + sugar → nucleoside; nucleoside + phosphate → nucleotide; nucleotide + nucleotide → nucleic acid

125. Choose the correct statements among the following
a) Habitat loss and fragmentation is the most important cause driving animals and plants to extinction.
b) Zoological parks and botanical gardens are in-situ conservation methods
c) The historic Convention "The World Summit' held in Rio de Janeiro in 1992,
d) 'Biodiversity hotspots' are the regions with high levels of species richness and high degree of endemism.
(1) a,b only (2) b,c,d
(3) a,c,d (4) a,d only

126. Match Column I with Column II and choose the correct option:

Column – I (Protein)		Column – II (Function)	
A.	Collagen	I.	Hormone
B.	Trypsin	II.	Intercellular ground substance
C.	Insulin	III.	Enzyme
D.	Antibody	IV.	Flights infectious agents

(1) A – II, B – III, C – I, D – IV
(2) A – III, B – II, C – IV, D – I
(3) A – II, B – I, C – III, D – IV
(4) A – IV, B – III, C – II, D – I

127. Select the correct combination of extinct animal.
(1) Thylacine - Africa
(2) Quagga - Mauritius
(3) Steller's sea cow – Russia
(4) Dodo – Australia

128. Select the incorrect statement from the following
(1) Fish diversity is higher than bird diversity.
(2) Columbia has less bird's diversity than India
(3) Constant and predictable environments promote niche specialisation and lead to a greater species diversity.
(4) In the species-area relationships among very large areas like the entire continents, the slope of the Z- line to be much steeper

129. Which of the following statements are incorrect?
I. In monocot stem, vascular bundles are arranged in a ring.
II. Phloem tissue is responsible for transport of water and minerals.
III. Angiosperms show double fertilisation.
IV. Xylem fibres provide mechanical support to plants.
V. Bryophytes require water for fertilisation.
Options:
(1) I and II (2) I, II and III
(3) II and IV (4) I, II and V

130. Match the following column-I with column – II and choose the correct answer from the key give below

Column – I	Column – II
A) Total lung capacity	i) TLC - RV
B) Function residual capacity	ii) VC + RV
C) Expiratory capacity	iii) TV + ERV
D) Vital capacity	iv) ERV + RV

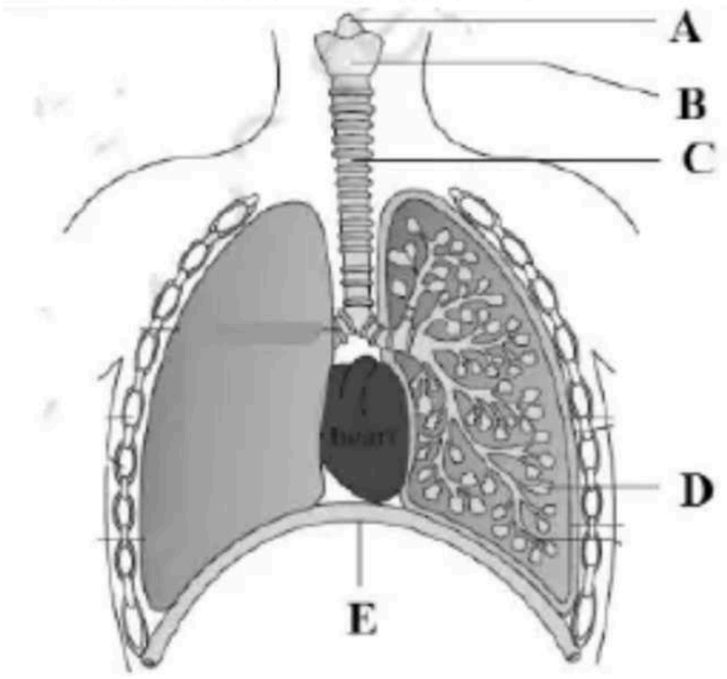
Choose the correct answer from the options given below:

(1) A-ii, B-iv, C-iii, D-i
(2) A-ii, B-iv, C-i, D-iii
(3) A-iv, B-ii, C-i, D-iii
(4) A-iv, B-ii, C-iii, D-i

131. Statement A: Formation of the enzyme–substrate (ES) complex is essential for enzyme catalysis.
Statement B: The enzyme undergoes a temporary conformational change after substrate binding, leading to product formation.

(1) Both A and B are true
(2) Both A and B are false
(3) A is true but B is false
(4) A is false but B is true

132. Observe the following diagram. With respect to A, B, C, D and E, Which of the following is the correct combination ?



(1) 'A' is cartilaginous but 'B' is supported with inter coastal muscles.
(2) 'C' is an un paired wind pipe. 'E' is dome shaped muscular structure.
(3) 'B' is sternum and 'C' is supported by cartilaginous rings
(4) 'A' is external nostril and 'E' supports heart as pericardium

133. Arrange the following structural features in the order of their appearance during Prophase I:
A. Fully condensed chromosomes with spindle formation
B. Homologous chromosomes paired as bivalents
C. Chiasmata visible between homologous chromosomes
D. Four chromatids visible as tetrads
E. Thin, long, thread-like chromosomes

(1) E → B → D → C → A
(2) E → D → B → C → A
(3) B → E → D → C → A
(4) E → B → C → D → A

134. What is the correct order of these events?
P) Conversion of fibrinogen to fibrin
Q) Clot retraction and leakage of serum
R) Thromboplastin formation
S) Conversion of prothrombin to thrombin

(1) R- Q- P- S (2) R- S -P –Q
(3) R -S -Q -P (4) S -P -R -Q

135. Activated sludge differs from primary sludge because activated sludge:

(1) Is rich in inorganic matter
(2) Contains mainly pathogenic microbes
(3) Consists of actively growing microbial flocs
(4) Is directly released into rivers

136. Given below are two statements:
Statement-I: Nodal tissue alone cannot regulate human heart function completely.
Statement-II: Though the heart is auto excitable, its functions can be moderated by neural and hormonal mechanisms.
In the light of the above statements, choose the correct answer from the options given below.

(1) Statement I is false but Statement II is true
(2) Both Statement I and Statement II are true
(3) Statement I is true but Statement II is false
(4) Both Statement I and Statement II are false

137. Match Column I with Column II and choose the correct option:

Column – I		Column – II	
A.	Carboxylation	I.	Formation of RuBP
B.	Reduction	II.	CO_2 fixation
C.	Regeneration	III.	Use of ATP and NADPH
D.	RuBisCo	IV.	Carboxylating enzyme

(1) A – IV, B – III, C – I, D – II

(2) A – III, B – II, C – I, D – IV

(3) A – II, B – I, C – III, D – IV

(4) A – II, B – III, C – I, D – IV

138. Statement A: Photorespiration does not occur in C_4 plants.

Statement B: C_4 plants increase CO_2 concentration around RuBisCO, favouring its carboxylase activity.

(1) Both A and B are true

(2) Both A and B are false

(3) A is true but B is false

(4) A is false but B is true

139. The functioning of the kidneys is efficiently monitored and regulated by hormonal feedback mechanisms involving

(1) Hypothalamus and pancreatic hormones

(2) JGA and thymus

(3) The heart and liver

(4) Hypothalamus, JGA and heart

140. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R :

Assertion (A) : Bony fishes excrete ammonia but kidneys do not play any important role in its removal.

Reason (R): Ammonia is the least toxic and readily soluble, so it is excreted by diffusion across body surface as ammonium ions.

In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both assertion and reason are true and reason is the correct explanation of assertion.

(2) Both assertion and reason are true but reason is not the correct explanation of assertion.

(3) Assertion is true but reason is false.

(4) Assertion is false but reason is true

141. Match the following

A	Hyoid bone	1	Part of rib cage
B	Mandible	2	Bone of pectoral girdle
C	Sternum	3	U-shaped bone
D	Scapula	4	Lower jaw

(1) A-1, B-3, C-2, D-4

(2) A-3, B-4, C-1, D-2

(3) A-3, B-1, C-2, D-4

(4) A-4, B-2, C-1, D-5

142. A leaf shows normal light reactions and adequate ATP and NADPH production, but carbon fixation rate is low despite sufficient light supply. Which condition best explains this?

(1) Absence of grana in chloroplast

(2) Low activity of Photosystem II

(3) High O_2 concentration competing with CO_2 at RuBisCO active site

(4) Defective electron transport chain

143. The successful use of retroviruses and *Agrobacterium tumifaciens* as gene delivery vectors is primarily based on their ability to:

(1) Replicate independently inside host cells without integration

(2) Kill host cells rapidly after infection

(3) Introduce their genetic material into host cells and alter host gene expression

(4) Produce antibiotics that select transformed cells

144. Which of the following is not accompanied with skeletal muscle contraction?
(1) The distance of two Z lines decreases
(2) Increased calcium level in sarcoplasm.
(3) Overlapping regions of actin and myosin reduces
(4) H zone decreases

145. Following statements regarding the central neural system are correct, except
(1) Forebrain consists of cerebrum, thalamus and hypothalamus.
(2) A canal called cerebral aqueduct passes through pons.
(3) Hypothalamus controls body temperature, eating and drinking
(4) Arachnoid is the middle layer of cranial meninges of human

146. Which of the following events does NOT occur during the Krebs cycle?
(1) Release of CO_2
(2) Reduction of NAD^+ to NADH
(3) Substrate level phosphorylation
(4) Direct utilisation of oxygen

147. Read the statements given below and select the correct one.
Statement-I: All the hormones of adrenal gland are steroid in nature.
Statement-II: The gastro intestinal hormones like Gastrin, Secretin, CCK, and GIP all of these are peptide hormones.
(1) Both statements I and II are correct.
(2) Both statements I and II are incorrect.
(3) Statement I is correct but statement II is incorrect.
(4) Statement I is incorrect but statement II is correct

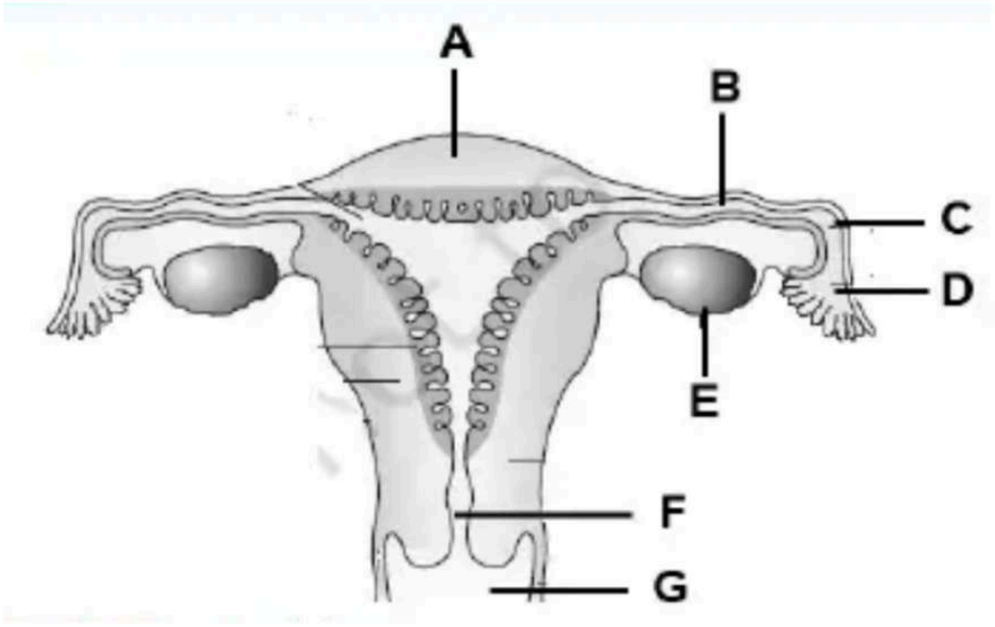
148. Which of the following statements regarding glycolysis is correct?
(1) It occurs only in mitochondria
(2) It requires oxygen for ATP production
(3) It is a universal pathway occurring in cytoplasm of all living cells
(4) It produces CO_2 as a by-product

149. Which of the following statements are correct?
I. Differentiation involves structural and functional specialisation of cells.
II. Dedifferentiation refers to mature cells regaining the ability to divide.
III. Dedifferentiation occurs when cells lose the ability to divide permanently.
IV. Development is the sum total of growth and differentiation.
V. All plant cells retain the ability to dedifferentiate.
(1) I, II and IV
(2) I, II, III and IV
(3) II, III and V
(4) I, III and IV

150. In the light of the above statements, choose the most appropriate answer from the options given below :
Assertion (A): Thyrocalcitonin and PTH have antagonistic effects.
Reason(R): Thyrocalcitonin is protein hormone but PTH is peptide hormone.
In the light of the above statements, choose the most appropriate answer from the options given below:
(1) Both Assertion and Reason are true and the reason is the correct explanation of the assertion.
(2) Both Assertion and Reason are true and the reason is not the correct explanation of the assertion.
(3) Assertion is true, but Reason is false.
(4) Assertion is false, but Reason is true

151. Removal of the apical bud followed by cytokinin application will most likely result in:
(1) Suppression of lateral buds
(2) Enhanced apical dominance
(3) Activation of lateral buds and bushy growth
(4) Early leaf senescence

152. Observe the labeled parts in the given diagram and pick out the correct option which indicates their function:



(1) A – is site for fertilization.

(2) B,C,D are parts of birth canal.

(3) F and G are parts of Oviduct.

(4) E produces female gametes and steroid hormones

153. Read the following statements :

Statement-I : Spermatogenesis starts at the age of puberty due to significant increase in the secretion of GnRH.

Statement-II : Oogenesis is initiated during embryonic development and completes its meiosis I and produces polar bodies before birth.

In the light of the above statements, choose the correct answer from the options given below :

In light of the above statements, choose the correct answer from the options given below.

(1) Both statement I and statement II are correct.

(2) Both statement I and statement II are incorrect.

(3) Statement I is correct but statement II is incorrect.

(4) Statement I is incorrect but statement II is correct

154. Which of the following statements correctly describes the role of origin of replication (ori) in a cloning vector?

(1) It helps in selection of transformed cells only

(2) It provides sites for restriction enzyme action

(3) It ensures expression of the foreign gene in the host

(4) It initiates replication of the inserted DNA and controls its copy number in the host

155. Which hormone/s attain peak level in the middle of menstrual cycle ?

(1) Progesterone and Estrogen

(2) Prolactin and hPL

(3) FSH and LH

(4) hCG and relaxin

156. A dark-grown dicot seedling shows shortened hypocotyl, exaggerated apical hook, and radial swelling. This response is best explained by:

(1) Increased auxin concentration

(2) Reduced gibberellin synthesis

(3) High ethylene concentration affecting cell elongation

(4) ABA-induced dormancy

157. Select incorrect statement about embryonic development.

(1) hPL and hCG are produced only during pregnancy.

(2) After implantation the inner cell mass differentiate into germ layers.

(3) Pregnancy lasts about 9 months in all mammals.

(4) First movements of foetus observed during fifth month

158. A hormonal IUD recommended to promote the cervix hostility to the sperms is

(1) Cu – T (2) Multiload – 375

(3) LNG – 20 (4) Lippes's loop

159. In a Mendelian dihybrid cross, 320 progeny are produced in F_2 generation. How many will show both recessive traits?

(1) 30 (2) 20

(3) 40 (4) 60

160. All the following statements are correct about "Saheli", but one is wrong. Which one is wrong ?

(1) New oral contraceptive pill for females

(2) Contains non-steroidal preparation

(3) It is once a week pill, developed by CDRI, Lucknow

(4) Single dose of this injection prevents pregnancy for years

161. What is the probability of obtaining TtRr genotype from a cross TtRr × TtRr?

(1) 1/14 (2) 1/4

(3) 3/16 (4) 1/16

162. Which of the following are the complications resulting from STIs without timely detection and treatment

A) Pelvic inflammatory disease (PID)

B) Still births

C) Infertility / cancer of reproductive tract

D) Ectopic pregnancies.

(1) A and B only (2) A, B and C only

(3) A only (4) A, B, C and D

163. Assertion (A): DNA fragments produced by the same restriction endonuclease can be easily joined to form recombinant DNA.

Reason (R): Restriction endonucleases generate complementary sticky ends that can hydrogen-bond with each other, facilitating the action of DNA ligase.

(1) Both A and R are true, and R is the correct explanation of A

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) A is false, but R is true

164. In a plant showing incomplete dominance, red (RR) × white (rr) flowers are crossed.

If 1,200 F_2 plants are obtained, how many plants will show phenotype identical to the F_1 generation?

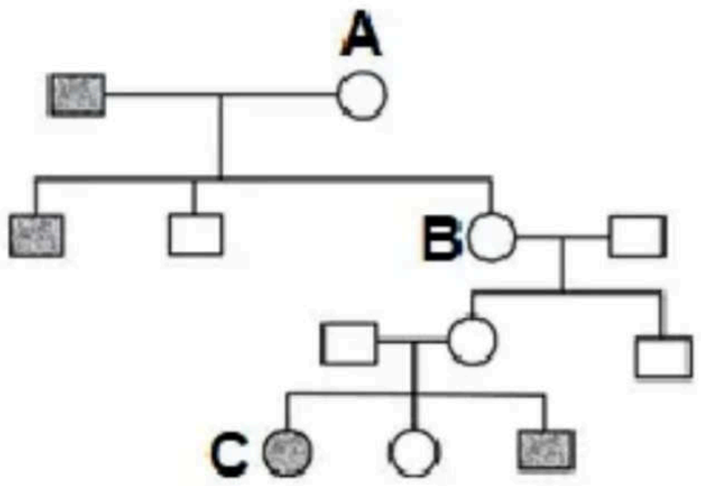
(1) 300 (2) 600

(3) 900 (4) 400

165. Select the correct difference between sickle cell anaemia and thalassemia

	Sickle cell anemia	Thalassemia
(1)	Quality of beta chain is affected	Quantity of alpha or beta chains affected
(2)	Quantitative problem	Qualitative problem
(3)	Controlled by two genes	Controlled by single gene
(4)	It is X linked recessive disease	It is autosomal recessive disease.

166. Inheritance of a Mendelian disorder is represented by the given pedigree chart. Identify the correct statement.



(1) It is X linked dominant disorder

(2) C is colorblind daughter

(3) It represents Phenylketonuria

(4) A and B are carriers for a dominant trait

167. Select the correct statement from the following

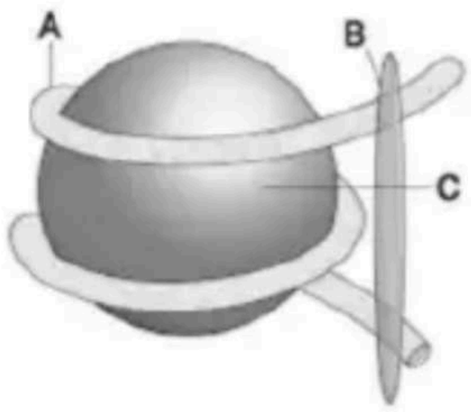
(1) VNTR are referred as Expressive Sequence Tags that express as RNA

(2) DNA from a single cell is enough to perform fingerprinting analysis.

(3) YAC and BAC vectors are used in hybridization of DNA fingerprints

(4) DNA fingerprinting is the basis for molecular diagnosis for early detection

168. Choose the correct identification of labels A, B and C.



(1) A – DNA, B – Histone octamer, C – H1 histone

(2) A – Histone octamer, B – DNA, C – H1 histone

(3) A – H1 histone, B – DNA, C – Histone octamer

(4) A – DNA, B – H1 histone, C – Histone octamer

169. Assertion (A): The unequivocal proof that DNA is the genetic material was provided by the experiments of Hershey and Chase.
Reason (R): They demonstrated that DNA, and not protein, enters the bacterial cell during bacteriophage infection and directs the synthesis of new viral particles.

(1) A is false, but R is true

(2) Both A and R are true, but R is not the correct explanation of A

(3) A is true, but R is false

(4) Both A and R are true, and R is the correct explanation of A

170. Match the correct combination.

(1) 65 mya - Dinosaurs suddenly disappeared from the earth.

(2) 500 mya - The first cellular forms of life appeared on earth.

(3) 350 mya - Invertebrates are formed and active.

(4) 200 mya - First jaw less fish evolved

171. Read the following statements (a–e).
Which of the following are incorrect?
a. Translation occurs in the cytoplasm of both prokaryotes and eukaryotes.
b. tRNA has an anticodon loop and an amino acid acceptor end.
c. UAA, UAG and UGA are stop codons.
d. Genetic code is overlapping in nature.
e. In bacteria 23S rRNA is associated with peptide bond formation.

(1) d only (2) d and e

(3) b and d (4) a, d and e

172. Transcription and translation can occur simultaneously in bacteria because:

(1) mRNA is unstable in bacteria

(2) Ribosomes are absent in nucleus

(3) There is no separation between cytoplasm and nucleus and mRNA requires no processing

(4) RNA polymerase directly interacts with ribosome

173. Which of the following is INCORRECT ?

(1) The fitness, according to Darwin, refers ultimately and only to reproductive fitness.

(2) Haldane and Oparin proposed that the first form of life could have come from pre-existing non-living organic molecules.

(3) Louis Pasteur demonstrated that the natural selection and branching descent are two key concepts of evolution

(4) Lamarck said that evolution of life forms had occurred but driven by use and disuse of organs

174. In the light of the above statements, choose the most appropriate answer from the options given below :
Assertion (A): Fitness is the end result of ability to adapt and get selected by nature.
Reason (R): The rate of appearance of new forms in evolution is linked to the life span or life cycle.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both A & R are correct and R is correct explanation of A.

(2) Both A & R are correct but R is not correct explanation of A.

(3) A is correct and R is incorrect

(4) A is incorrect but R is correct

175. The peptidyl transferase activity of ribosome is due to:

(1) Ribosomal proteins of large subunit

(2) Small subunit rRNA

(3) ATP hydrolysis by ribosome

(4) 23S rRNA acting as a ribozyme

176. The lac operon is switched ON in the presence of lactose because:

(1) Lactose binds directly to RNA polymerase

(2) Repressor protein is not synthesised

(3) Inducer inactivates the repressor allowing transcription

(4) Operator region is removed

177. Match the contents of column I with column II and choose correct answer.

List – I	List – II
(A) α - thalassemia	(i) Transgenic cow
(B) α -1 antitrypsin	(ii) Emphysema
(C) α - interferons	(iii) Genetic disorder
(D) α -Lactalbumin	(iv) Cancer treatment

(1) A – iii, B – ii, C – iv, D – i

(2) A – iii, B – i, C –ii, D – iv

(3) A – iii, B – i, C – iv, D – ii

(4) A – i, B – ii, C – iv, D – iii

178. Read the following statements and choose the correct statements.

(a) ELISA is a nucleic acid based diagnostic tool that can confirm presence of infectious microbe at early stages.

(b) In general, insulin from r DNA technology does not show any side effects.

(c) Injecting ADA enzyme directly in to the patient with it's deficiency can be completely curable.

In the above statements the true statements are

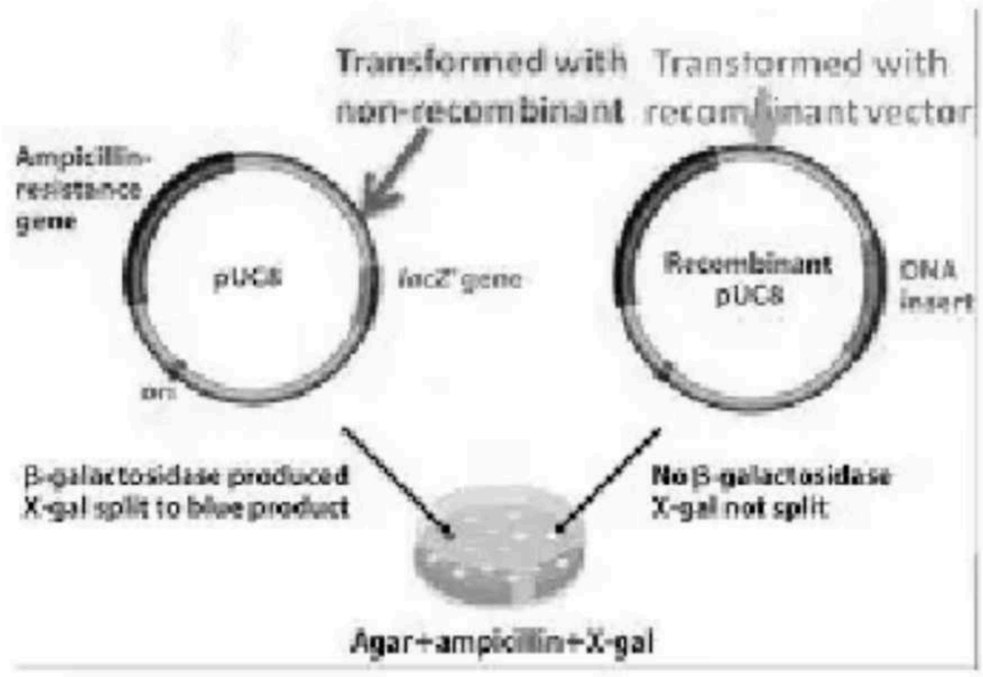
(1) (a) and (c) only

(2) all the (a), (b) and (c)

(3) only (b)

(4) (a) and (b) only

179. The diagram shows transformation of *E. coli* with pUC vector and screening on agar containing ampicillin and X-gal.



Which of the following statements is correct?

(1) Blue colonies indicate insertion of foreign DNA at the lacZ gene

(2) White colonies lack plasmid and are sensitive to ampicillin

(3) White colonies represent recombinants due to insertional inactivation of lacZ gene

(4) Blue colonies are recombinants and do not produce β-galactosidase

180. Read the following statements carefully and choose the correct option.

Statements:

A. Lysozyme is used to break open bacterial cells during DNA isolation.

B. Cellulase is required for breaking animal cell membranes to release DNA.

C. Histone proteins associated with DNA are removed using protease.

D. RNA is removed from the DNA preparation by treatment with ribonuclease.

E. Purified DNA precipitates out on addition of chilled ethanol and appears as fine threads.

Which of the following statements are correct?

(1) A, C, D and E

(2) A, B, C and D

(3) B, C and E only

(4) A, D and E only